Querying RMBS trustee data in natural language, prototype results

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A language model converts each question into a structured query spec, and a compiler turns the spec into SQL using metric definitions held in code, so the model never does the arithmetic. These results come from a working prototype on synthetic data, covering 50 Australian RMBS deals, 21,945 loans and 773,336 loan-months, in 2 trustee formats.

Test results

test	result
Cleaned table against simulated truth	0 differences across 773,336 loan-months
Resubmitted duplicates removed	721 of 774,057 raw rows
Clean-up calls reported as repayments, recovered	7 deals
Compiler output against hand-written SQL	49 of 49 match, worst difference 1.6e-15
Compiled queries valid for Oracle	49 of 49
Slowest question at 27,840,096 loan-months	under 0.75 s, default threads

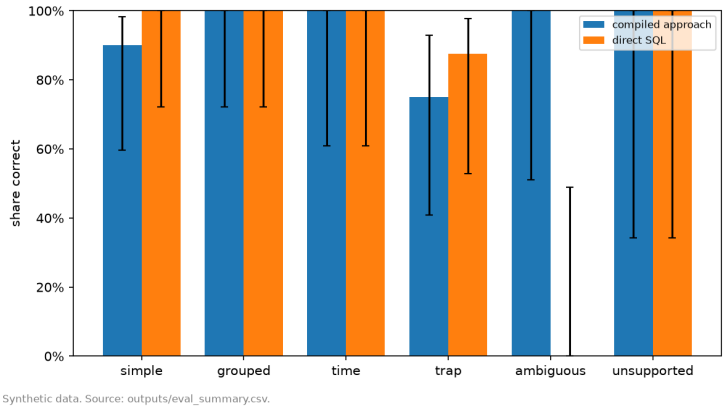
What common mistakes do to the numbers

mistake	question	correct	with the mistake
2 missed payments read as 60+ days	Q27	0.66%	0.59%
Clean-up calls counted as prepayments	Q29	21.6%	100.0%
Clean-up calls counted as prepayments	Q30	18.3%	49.0%
Exit month counted as a live loan	Q32	9,975	10,572
Simple average, not balance-weighted	Q33	51.96%	50.92%
Monthly CPR averaged, not compounded (dropped)	Q2 2025 CPR	18.064%	18.063%

Compared with direct SQL

Same 40 questions, with the same table and metric definitions in the direct SQL model's prompt. Neither approach returned an incorrect number, and every miss on both sides was a refusal to answer. On ambiguous questions, the compiled approach asked what was meant, where direct SQL declined without explanation.

category	questions	compiled approach	direct SQL
simple	10	9 (90%)	10 (100%)
grouped	10	10 (100%)	10 (100%)
time	6	6 (100%)	6 (100%)
trap	8	6 (75%)	7 (88%)
ambiguous	4	4 (100%)	0 (0%)
unsupported	2	2 (100%)	2 (100%)
total	40	37 (92%)	35 (88%)



Synthetic data. Source: outputs/eval_summary.csv.

Asked "What are arrears in NSW?", the compiled approach replied with a question.

```
Which arrears measure do you want for NSW?
Options: 30+ days past due; 90+ days past due
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